

Lesson 15 Strong Induction

In Lesson 14, we learned the technique of induction to prove a statement $S(n)$ of all integers $n \geq a$. In this lesson, we introduce the **strong** version of induction.

- Basis step: prove that $S(n)$ is true for base case(s) (may be more than one)
- Inductive step: for any $k \geq a$, assume that $S(n)$ is true for all $a \leq n \leq k$ and show that $S(k + 1)$ is true.

Strong induction has a *stronger* induction hypothesis than “regular” induction, that is, we assume more than just $S(k)$ is true, we assume that $S(a), S(a + 1), \dots, S(k)$ are all true. This is necessary in some proofs as we will see.

Fundamental Theorem of Arithmetic

Every integer $n > 1$ has a prime factorization, that is, $n = p_1 p_2 \dots p_k$, where $p_1, p_2, \dots, p_k$ are primes.

Proof The statement is true for $n = 2$: 2 is prime, so $2 = 2$ is a product of primes. Suppose that the statement is true for $2, 3, 4, \dots, k$, that is, they can all be written as a product of primes. Now $k + 1$ is either a prime number or it is not.

If $k + 1$ is prime, then $k + 1$ is a product of primes.

If $k + 1$ is not prime, then $k + 1 = mr$, where $m, r \in \mathbb{Z}$ and $1 < m, r < k + 1$.

By the induction hypothesis, m and r can be written as products of primes, say

$$m = p_1 p_2 \dots p_j, \quad r = q_1 q_2 \dots q_l$$

where p_i 's and q_i 's are primes.

So $k + 1 = mr = p_1 \dots p_j q_1 \dots q_l$ is a product of primes, and $k + 1$ has a prime factorization, and by induction, every integer $n > 1$ has a prime factorization.

Proposition 1 Consider the recursive sequence:

$$b_1 = 5, b_2 = 13, \text{ and } b_n = 5b_{n-1} - 6b_{n-2} \text{ for } n \geq 3$$

Then for all $n \in \mathbb{N}$, $b_n = 2^n + 3^n$.

Proof (Basis step) We have two base cases $n = 1$ and $n = 2$ (the recursive definition only starts for $n \geq 3$), and we need to verify that:

$$b_1 = 2^1 + 3^1 = 5$$

$$b_2 = 2^2 + 3^2 = 13$$

(Inductive step) Let $k \geq 2$ be a natural number, we assume that $b_n = 2^n + 3^n$ for all natural numbers $n \leq k$. Then

$$\begin{aligned} b_{k+1} &= 5b_k - 6b_{k-1} \\ &= 5(2^k + 3^k) - 6(2^{k-1} + 3^{k-1}) \\ &= 5(2^k + 3^k) - (2 \cdot 3)(2^{k-1} + 3^{k-1}) \\ &= 5(2^k) + 5(3^k) - 3(2^k) - 2(3^k) \\ &= 2(2^k) + 3(3^k) = 2^{k+1} + 3^{k+1} \end{aligned}$$

Our next example involves the **Fibonacci numbers**, defined recursively as follows:

$$F_1 = 1, \quad F_2 = 1, \quad F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 3$$

The first few Fibonacci numbers are 1, 1, 2, 3, 5, 8, 13, ... We will prove that any positive integer can be expressed as a sum of distinct Fibonacci numbers. But we first prove a few lemmas (small facts which help to prove the main theorem).

Lemma 1 All Fibonacci numbers are positive.

Proof We can prove this simple fact using strong induction.

(Basis step) This is true for $n = 1$ and $n = 2$: $F_1 = F_2 = 1 > 0$.

(Inductive step) For an arbitrary $k \geq 2$, suppose $F_n > 0$ for all $1 \leq n \leq k$. Then

$$F_{k+1} = F_k + F_{k-1} > 0$$

Lemma 2 Let F_n be the Fibonacci sequence. Then $F_n < F_{n+1} \leq 2F_n$ for all $n \geq 2$.

Proof For $n \geq 2$, we have $F_{n+1} = F_n + F_{n-1} > F_n$ since $F_{n-1} > 0$ (Lemma 1).

For $n = 2$, $F_3 = 2 \leq 2F_2 = 2(1)$.

For $n \geq 3$, we have $F_{n+1} = F_n + F_{n-1} < F_n + F_n = 2F_n$ since $F_{n-1} < F_n$.

Proposition 2 Every positive integer can be written as a sum of distinct Fibonacci numbers.

Proof Since $F_1 = F_2 = 1$, use only F_l , $l \geq 2$, in a sum of distinct Fibonacci numbers.

(Basis step) The statement is true for $n = 1$: $1 = F_2$.

(Inductive step) Suppose for any $k \geq 1$, all positive integers $n \leq k$ can be expressed as a sum of distinct Fibonacci numbers. If $k + 1$ is a Fibonacci number, we are done. If not, we write:

$$k + 1 = F_j + m$$

where F_j is the largest Fibonacci number less than $k + 1$. We will show that $m < F_j$ by contradiction. Suppose $m \geq F_j$, then

$$k + 1 = F_j + m \geq F_j + F_j = 2F_j \geq F_{j+1}$$

Since $k + 1$ is not a Fibonacci number, $k + 1 > F_{j+1} > F_j$, contradicting that F_j is the largest Fibonacci number less than $k + 1$.

Since $0 < F_j < k + 1$, we have $0 < m < k + 1$, and m can be written as a sum of distinct Fibonacci numbers.

Since $m < F_j$, m as a sum of distinct Fibonacci numbers does not include F_j . This implies that $k + 1 = F_j + m$ is a sum of distinct Fibonacci numbers.